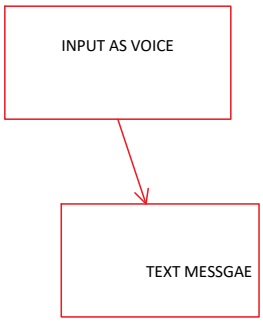
RAG :-
RETRIVAL-AGUMENTED GENERATION,A TECHNIQUE IN AI (SPECIALLY IN NATURAL LANGUAGE PROCESSING) , THAT IMPROVES HOW MODELS LIKE , CHABOTS ANSWER QUESTIONS …



RAG= Search + generate
It combines 2 steps :-

1. **Retrieval:-the system will look for relevant information from a database or documents**
2. **Generation:- Ai language model(like Open ai, GPT model) uses that retrieved information to produce a more accurate and context-aware answer.**

Why RAG useful:-
Normally, an AI model only knows what it learned during the training ….
Used RAG:-
It can access up-to-date information
It can answer questions about specific documents or private data
It reduces hallucinations (imaginations, assumptions) made up answers
Example:-
what does my company internal policy says about remote work?
A standard AI:- doesn't know your company policy
A RAG system:- it going to retrieve your company policy document then explains it

Where its used :-

- **Chat boards that answers from the document**
- **Search engine with AI summaries**
- **Customer support system**
- **Knowledge assistance for companies**

How the RAG works :-
Step 1:- user ask question
Example - what are the refund rules in company policy ?

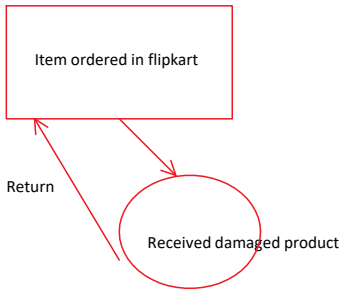
Step2:-convert the question in to embedding's

The question is transformed into numeric representation(vector) so the system can compare the meaning ..not just a keyword

Step3:-retrieve relevant document

The system searches a knowledge /vector database finds the most relevant chunks of text, such as :-

- Refunds allowed within 30days with receipt;
- Digital purchase are non-refundable



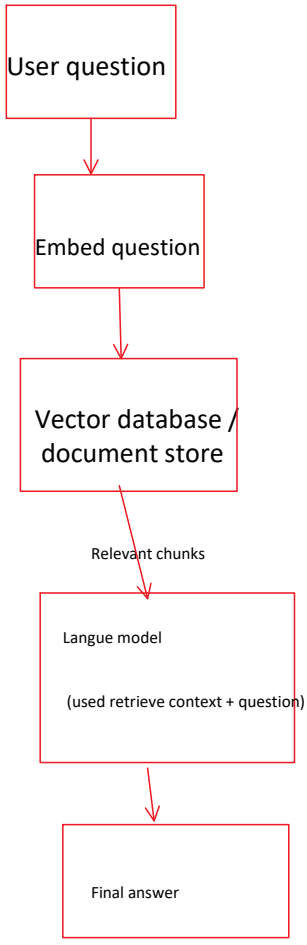
Step4:-build a prompt with context

Those retrieved passages are instead in to the model prompt along with user's question…

Prompt is :

Use the following company policy generate answers

- Refunds allowed within 30days with receipt;
- Digital purchase are non-refundable
- Questions:-what are the refund rules in company policy ?



Step5:- generate the answer

The language model uses the retrieved content to write grounded response ;-

Example- refunds allowed within 30days with receipt , but digital purchases excluded

Common tools used into RAG System:-

- Embedding model
- Vector Database(pinecone, weaviate ,milvus)
- LLMs like OpenAI model, Gpt models
- Frameworks like langchain , or LlamaIndex

Vector Database:-

A vector database is a specialized database designed to , store, index, and search vector embedding numerical representation of data like ,text,images,audios,videos

Why vector database useful?

In traditional database (sql,nosql) get exact match

Select * from emp where Empid=104

vector database is great for semantic questions

"find similar article to this article"

SQL database

EmpID	EMP_Name	salary
101	Murali	50000
102	Kumar	20000
103	Raju	10000
104	Ramya	52000
105	Krishna	32000

Select * from emp where salary>30000

Key features :-

"find similar article to this article"

Key features :-
- Similarity search
- High-dimensional indexing (vector often have hundreds m thousands, of dimensions)
- Fast retrieval even with million or billion off vectors
- Often used with AI/ML pipelines

Common use cases ;
- Semantic search (search by meaning)
- Recommendation systems
- Chabot's and RAG
- Image and video searching
- Fraud detection and anomaly detection
-
-

Vector database tools :-
- Pinecone --production SaaS
- Weaviate--> millions of docs + filters
- Milvus--> billions of embeddings
- FAISS-->custom in-house Ml infra
- Chroma --. Fast prototype--

Agentic AI:-

Agent based artificial intelligent ;
It refers to AI system that can act independently to achieve goals, rather than just responding to prompts

Instead of being passive tools(chatboat answering questions),agentic AI behaves more like a digital "agent" that can :-
- Make decisions based on context
- Plan action to reach the goal '
- Takes the steps automatically (call API, Use tools ,run code)
- Adapt based on feedback or results

A normal AI :-
Write an email ..> it will write an email
An agentic AI:-
You say ," schedule a meeting with hr person name as kiran in next "
1.check the calendar
2.find free timeslots
3.email to kiran
4.confirm the meeting
5.add it in to your calendar

These steps are done without your instructions

Key characteristics of Agentic AI:-

- Goal(what they are trying to achieve)
- Memory(tract past actions or context)
- Planning ability (break the tasks in to steps)
- Tools Used(APIs, Database, APPs)
- Autonomy (operate with minimal human input)

Where its used?;;--

Select * from emp where salary>30000

101	Murali	50000
104	Ramya	52000
105	Krishna	32000

Select * from emp where Empid= 104

104	Ramya	52000

- Person assistance (advanced version of siri, alexa etc)
- Autonomous software bots(customer support, scheduling)
- Robotics and self-driving systems
- AI workflows in business (automating the tasks end-to-end)

1. Customer Support AI agents(amazon support bots handling returns automatically.)
2. AI coding agents (github github copilot ,autonomous dev agents)
3. Self-driving cars (autonomous vehicle ,tesla autopilot , waymo robotaxis.)
4. Personal AI assistance
5. Finance / banking AI agents … credit card fraud detection system acting instantly
6. Healthcare AI agents can be used
7. Cybersecurity AI agents used

Top 10 agentic AI companies in 2026--;

1. OpenAI
   a. Ai agents for coding ,workflows, enterprise automations
2. Google/Google DeepMind ;
   a. Gemini-based entreprised AI agents
3. Microsoft
   a. Copilot ecosystem
4. Anthropic :
   a. Claud-powered enterprise agents
5. Amazon /Amazon Web services
   a. To provide infrastructure for large-scale worloads
6. Salesforce - CRM- focused application
7. serviceNow- internal enterprise workflows
8. Uipath -> agentic automation
9. Nvidia -> infrastructure+internal deployments of AI coding agents
10. DeepSeek
    a. Rapid progress in open models , agentic coding capability

Claude Ai :-

Claude ai is generative AI assistance developed by Anthropic . It is designed to help with writing , coding , summarizing, research, reasoning, conversation.

First introduced in 2023

Features of Claude AI:-

- Natural conversations
- Writing essay , email reports
- Coding assistance
- Summarizing the long documents
- Answering the questions
- Brainstorming ideas
- Image understanding

Claude Model versions:-

It is having several generations , in that common families includes:,

1. Haiku -fast and lightweight
2. Sonnet- balance speed + intelligence
3. Opus -most powerful for complex tasks